

Quantum Teleportation Experiment Report

1. Objective

This report describes a complete implementation of the standard single-qubit quantum teleportation protocol using pyqpanda3. The state to be teleported is $|\psi\rangle = \cos(\pi/6)|0\rangle + \sin(\pi/6)|1\rangle$. The protocol transfers this unknown state from Alice to Bob by using a shared Bell pair, Alice's joint measurement, two classical bits, and Bob's conditional correction gates.

2. Bell-state preparation

The function `prepare_bell_state(q1, q2)` returns a pyqpanda3 QCircuit. It first applies $H(q1)$, which transforms $q1$ into $(|0\rangle + |1\rangle) / \sqrt{2}$. It then applies $CNOT(q1, q2)$. The resulting two-qubit state is $(|00\rangle + |11\rangle) / \sqrt{2}$, where $q1$ is Alice's entangled qubit and $q2$ is Bob's entangled qubit.

3. Teleportation circuit construction

The function `teleportation_circuit(q0, q1, q2)` builds the whole protocol.

Step 1: prepare the unknown state on $q0$. In pyqpanda3, $RY(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$. Therefore $\theta = \pi/3$ prepares $\cos(\pi/6)|0\rangle + \sin(\pi/6)|1\rangle$.

Step 2: call `prepare_bell_state(q1, q2)` to create the shared Bell state.

Step 3: Alice applies $CNOT(q0, q1)$ and $H(q0)$, then measures $q0$ into classical bit $c0$ and $q1$ into classical bit $c1$.

Step 4: Bob applies $X(q2)$ if $c1 = 1$ and applies $Z(q2)$ if $c0 = 1$. These conditional corrections are implemented with pyqpanda3 qif classical-control blocks.

4. Fidelity verification method

After running the circuit on CPUQVM, the script reads the final state vector. pyqpanda3 uses little-endian indexing for state-vector basis states, so the code extracts Bob's one-qubit state from the bit position of $q2$. For pure states, the fidelity is computed as $F = |\langle\psi|\phi\rangle|^2$, where $|\psi\rangle$ is the original state and $|\phi\rangle$ is Bob's recovered state after correction. A global phase is removed before displaying amplitudes, because it has no physical effect.

5. Numerical results

Original amplitudes: $\alpha = 0.866025403784$ and $\beta = 0.500000000000$.

Recovered Bob state after a full teleportation run: $0.866025403784|0\rangle + 0.500000000000|1\rangle$.

Recovered Bob probabilities: $P(0) = 0.750000000000$ and $P(1) = 0.250000000000$.

Single-shot state fidelity: $F = 1.000000000000$.

6. Branch-by-branch analysis

Alice bits	Branch probability	Bob state after correction	Fidelity
00	0.250000000000	$0.866025403784 0\rangle + 0.500000000000 1\rangle$	1.000000000000
01	0.250000000000	$0.866025403784 0\rangle + 0.500000000000 1\rangle$	1.000000000000

10	0.250000000000	$0.866025403784 0\rangle + 0.500000000000 1\rangle$	1.000000000000
11	0.250000000000	$0.866025403784 0\rangle + 0.500000000000 1\rangle$	1.000000000000

7. Conclusion

All four Alice measurement outcomes occur with probability $1/4$. In every branch, the corresponding X and Z corrections transform Bob's qubit back to the original state. The calculated fidelity is 1.0 for every branch, which confirms that the pyqpanda3 implementation successfully realizes the quantum teleportation protocol.